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Comparison between ensilage and fungal pretreatment for storage of giant reed and subsequent methane production

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HIGHLIGHTS

- Ensilage enhanced biomass preservation and methane yield of giant reed.
- Fungal pretreatment reduced glucose yields of giant reed harvested in August and October.
- Fungal pretreatment decreased methane yield of giant reed despite of harvest time.
- Ensilage was more suitable than fungal pretreatment for giant reed storage and AD.

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ABSTRACT

Ensilage and fungal pretreatment of giant reed harvested from August through December were compared based on their effects on feedstock preservation, glucose yield, and subsequent methane production via anaerobic digestion (AD). Compared to fungal pretreatment, ensilage obtained lower total solids (<1.2%) and cellulose (<3.5%) losses, and comparable hemicellulose degradation, except for giant reed harvested in August. Ensilage increased glucose and methane yields by 7–15% and 4–14%, respectively, for giant reed harvested from August through December. Fungal pretreatment failed for giant reed harvested in August and October with reduced glucose yields, and was effective for that harvested in November and December, with about 20% increases in glucose yield. However, hydrocarbon losses during fungal pretreatment offset the increased glucose yield, resulting in decreased methane yields by AD. In summary, ensilage was found to be more suitable than fungal pretreatment for giant reed storage and its methane production via AD.

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1. Introduction

Renewable energy production from biomass has the potential to reduce our dependence on fossil fuels, and mitigate climate change due to greenhouse gas emissions (World Energy Council, 2013). Since the production of biomass is usually seasonal with a limited window for harvest, storage of biomass is crucial for maintaining a viable bioenergy supply chain. Dry storage is the most commonly used method for storing agricultural biomass, especially for low-moisture crops. In order to save costs, agricultural biomass is typically dried in the field without artificial drying; however, unpredictable weather conditions present serious challenges (Sultana and Kumar, 2011). Dry storage also has other drawbacks such as high dry matter losses during outdoor storage and risk of fire (Shinners et al., 2007). In contrast, wet storage with concurrent microbial pretreatment, such as ensilage and fungal pretreatment, have been proposed to reduce dry matter loss and/or reduce biomass recalcitrance for improved bioenergy production (Cui et al., 2012; Liu et al., 2015).

Ensilage, a traditional process for preserving green crops in the livestock industry, has been considered as an effective and reliable technology for biomass storage (Herrmann et al., 2011). Ensilage usually relies on naturally existing microorganisms, mainly lactic acid bacteria, to convert water soluble carbohydrates (WSC) in the biomass to organic acids, such as lactic acid, acetic acid, propionic acid, and butyric acid. The production of organic acids reduces pH values in the silage to below 4, which inhibits the growth of microorganisms and thus preserves biomass (Weiland, 2010). The dry matter loss can be as low as 1–5% after one year of ensilage (Herrmann et al., 2011), and the ensiled biomass was also found to be more digestible than that from dry storage (Cui et al., 2012). The

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the intermediate products are completely degraded to CO_2. The fungal growth also consumes lignin, hemicellulose, WSC, and other carbohydrates. Lignin degradation is an important indicator for the success of fungal pretreatment. In this study, fungal pretreatment failed for giant reed harvested in August and October, with almost no or minimal lignin degradation, but was effective for giant reed harvested in November and December with lignin degradations of up to 7.9% (Fig. 1b). Similar results were reported by Vasco-Correa and Li (2015) who observed minimal lignin degradation in *Miscanthus* harvested in the fall, but more than 25% degradation in that harvested in the spring. Two possible reasons have been suggested for this observation, one is the higher nitrogen content of giant reed in the fall, which has been shown to influence the enzymatic activity of ligninolytic enzymes (Tekere et al., 2001); the other is the presence of antifungal compounds in the extractives, because hot water extraction was found to enable the growth of *C. subvermispora* on soybean stalks (Wan and Li, 2011a). Further investigations are needed to determine the reason for the failure of the fungal pretreatment on early harvested giant reed.

3.4. Effect of ensilage and fungal pretreatment on glucose yield of giant reed

The glucose yield of untreated giant reed decreased about 14% when the harvest date was delayed from August to December (Fig. 2), which may be attributed to the increased biomass recalcitrance as indicated by the higher lignin content (Table 1). Ensilage substantially increased glucose yield by 7.6%, 12.9%, 18.3%, and 20.3% for giant reed harvested in August, October, November and December, respectively (Fig. 2). The increased glucose yield during ensilage should be attributed to the lignin and hemicellulose degradation in giant reed, despite more than 8% cellulose loss during storage (Fig. 1b and c). Improved sugar yield by ensilage has also been reported in previous studies but with different feedstocks (Oleskowicz-Popiel et al., 2011; Zheng et al., 2011).

Different from ensilage, fungal pretreatment decreased glucose yields of giant reed harvested in August and October by 21% ($p < 0.05$) and 4% ($p > 0.05$), respectively. The reduced glucose yield of giant reed harvested in August and October indicated the failure of the fungal pretreatment, which was consistent with their low lignin degradation as well as the high cellulose loss for giant reed harvested in August (Fig. 1b and c). Fungal pretreatment increased the glucose yields of giant reed harvested in November and December by 20% and 22%, respectively, which was also consistent with their significant lignin degradation ($p < 0.05$) (Fig. 1b).

As shown in Fig. 2, ensilage was more effective than fungal pretreatment in improving glucose yield from giant reed harvested in August and October. However, fungal pretreatment resulted in comparable ($p > 0.05$) or significantly higher ($p < 0.05$) glucose yields than ensilage for giant reed harvested in November and December, respectively. Fungal pretreatment may be more suitable than ensilage for glucose-based bioenergy production, such as ethanol fermentation, from giant reed harvested in December. However, high glucose yield is not necessarily associated with high methane yield by AD, as AD also utilizes other organic compounds such as organic acids and hemicellulose. As a result, the effect of ensilage and fungal pretreatment on methane production by AD was further evaluated as described in the next section.

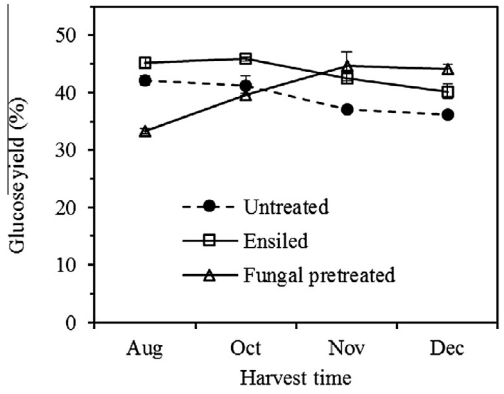

Fig. 2. Effect of ensilage and fungal pretreatment on glucose yield of giant reed harvested at different times.

3.5. Comparison of methane production from ensiled and fungal pretreated giant reed

As shown in Fig. 3, during AD of untreated, ensiled and fungal pretreated giant reed, the daily methane yield peaked on day 2–5, and most of the methane was produced within 15 days. Methane contents of biogas from ensiled giant reed and fungal pretreated giant reed increased quickly to 61–62% and 63–66% in 4 days during AD, while untreated giant reed produced more CO_2 at the initial stage of AD and its methane content reached 60–61% in 8 days (data not shown).

Ensiled giant reed had higher daily methane yields compared to those of untreated giant reed. Peak daily methane yields of ensiled giant reed harvested in August, October, November and December were 31.5%, 43.5%, 44.3%, and 10.6% higher than those of untreated giant reed, respectively (Fig. 3). The increased peak values might be related to the large amounts of organic acids and ethanol produced during ensilage, since these organic compounds are easily digestible substrates for AD, and accounted for 4–8% of the TS in ensiled giant reed (Table 2). This result was also consistent with a previous study that showed a good correlation between methane yield and the concentration of acetic acid and the total amount of acetic acid, lactic acid, and ethanol (Liu et al., 2015). Besides, the recalcitrance of giant reed was reduced by ensilage (Fig. 2), which could also have contributed to the increased peak values. In contrast to ensilage, fungal pretreatment negatively affected methane production from giant reed, resulting in 26.5–39.4% lower peak daily methane yields than the untreated giant reed harvested at different times (Fig. 3), probably due to its failure to reduce the biomass recalcitrance and/or loss of some easily degradable hydrocarbons during fungal growth (Vasco-Correa and Li, 2015).

The 30-day cumulative methane yields during AD of untreated giant reed decreased gradually from 165 L/kg VS to 141 L/kg VS when the harvest was delayed from August to December, although no significant ($p > 0.05$) difference was found between that harvested in August, October, and November (Fig. 4). Compared to untreated giant reed, ensilage resulted in a 13.7% and 10.2% increase ($p < 0.05$) in cumulative methane yields for giant reed harvested in August and December, respectively, and a slightly increased (about 5%) cumulative methane yield for giant reed harvested in October and November (Fig. 4). Fungal pretreated giant reed obtained 33.1%, 25.8%, 21.6%, and 7.6% lower cumulative methane yields than untreated giant reed harvested in August, October, November, and December, respectively (Fig. 4).

During ensilage and fungal pretreatment, some AD substrates such as cellulose, hemicellulose, WSC, and extractives were degraded, which negatively affected the methane yield. However, organic acids and/or ethanol were also produced especially during the ensilage process, which positively affected the methane yield. Besides, degradation of lignin and hemicellulose also had positive effects on methane production due to reduced biomass recalcitrance. As a result, the overall effect of ensilage and fungal pretreatment on the subsequent methane yield by AD was a trade-off between negative and positive effects.

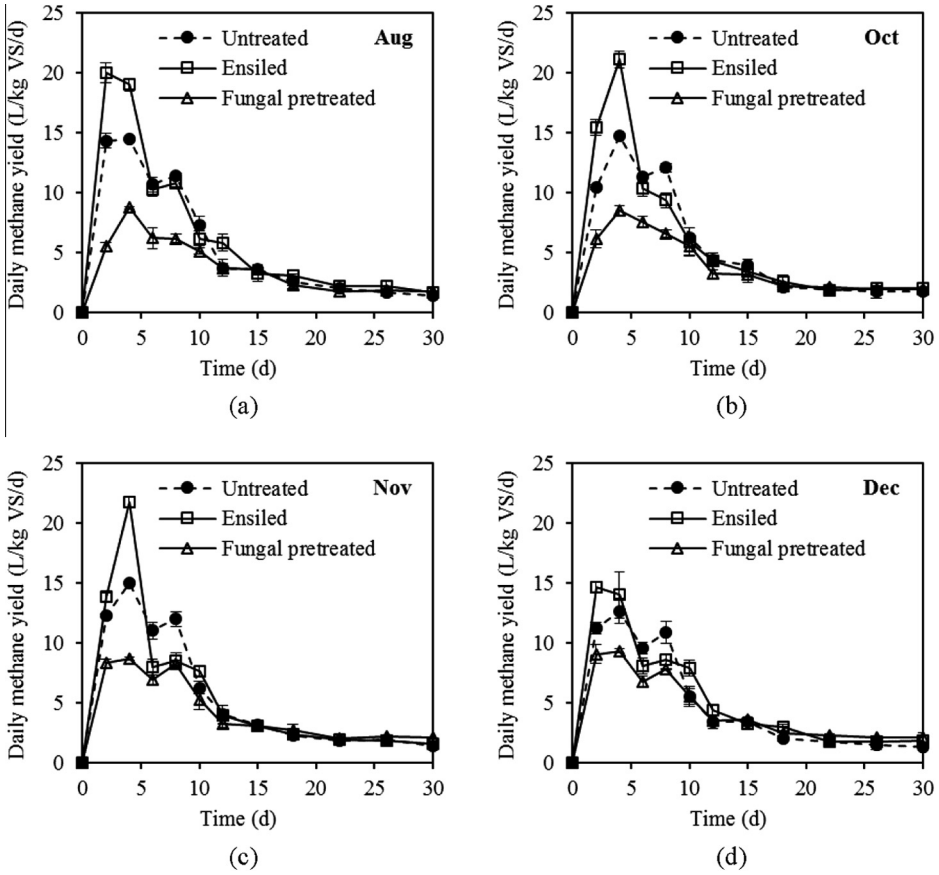

Fig. 3. Effect of ensilage and fungal pretreatment on daily methane yield during AD of giant reed harvested in August (a), October (b), November (c), and December (d).

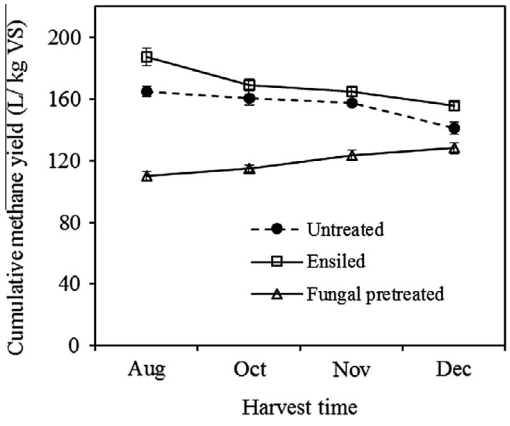

Fig. 4. Effect of ensilage and fungal pretreatment on cumulative methane yield during AD of giant reed harvested in August, October, November, and December.

As shown in Fig. 1, fungal pretreatment resulted in 5–7% reduction of hemicellulose, 25–30% reduction of extractives, and 60% reduction of WSC, in giant reed harvested in November and December (Fig. 1d–f). One previous study also reported decreased methane yield by fungal pretreatment with *C. subvermispora*, despite its effectiveness in improving sugar yields (Vasco-Correa and Li, 2015). Thus, the reduction of hydrocarbons during fungal pretreatment can offset the positive effects of reduced biomass recalcitrance, resulting in decreased cumulative methane yields during AD (Fig. 4). Although the ensilage process was shown to cause higher reduction of extractives and WSC than fungal pretreatment (Fig. 1e and f), these components were not totally lost but were converted to organic acids and ethanol that could be easily digested for methane production (Liu et al., 2015). However, the organic matter degraded during fungal pretreatment was mostly converted to CO_2, and thus could not contribute to the methane production. Therefore, ensiled giant reed can result in higher methane yields than untreated and fungal pretreated giant reed harvested at different times (Fig. 4).

4. Conclusion

Ensilage was better than fungal pretreatment for preserving giant reed harvested from August through December. Ensilage also reduced the recalcitrance of giant reed and improved its glucose and methane yields for all harvested times. In contrast, fungal pretreatment was only effective in improving glucose yields for giant reed harvested in November and December, and it resulted in reduced methane yields for giant reed harvested in August through December. Generally, ensilage was more suitable than fungal pretreatment for wet storage of giant reed for subsequent biogas production via AD.

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References

APHA, 2005. Standard Methods for the Examination of Water and Wastewater. American Public Health Association, Washington, D.C..

Cui, Z., Shi, J., Wan, C., Li, Y., 2012. Comparison of alkaline-and fungi-assisted wet-storage of corn stover. Bioresour. Technol. 109, 98–104.

Di Nasso, N.N., Angelini, L.G., Bonari, E., 2010. Influence of fertilisation and harvest time on fuel quality of giant reed (*Arundo donax* L.) in central Italy. Eur. J. Agron. 32, 219–227.

Di Nasso, N.N., Roncucci, N., Bonari, E., 2013. Seasonal dynamics of aboveground and belowground biomass and nutrient accumulation and remobilization in giant reed (*Arundo donax* L.): a three-year study on marginal land. Bioenergy Res. 6, 725–736.

Di Nasso, N.N., Roncucci, N., Triana, F., Tozzini, C., Bonari, E., 2011. Seasonal nutrient dynamics and biomass quality of giant reed (*Arundo donax* L.) and miscanthus (Miscanthus x giganteus Greef et Deuter) as energy crops. Ital. J. Agron. 6, e24.

Driehuis, F., Elferink, S.J.W.H., Spoelstra, S.F., 1999. Anaerobic lactic acid degradation during ensilage of whole crop maize inoculated with *Lactobacillus buchneri* inhibits yeast growth and improves aerobic stability. J. Appl. Microbiol. 87, 583–594.

Elferink, S.J.O., Krooneman, J., Gottschal, J.C., Spoelstra, S.F., Faber, F., Driehuis, F., 2001. Anaerobic conversion of lactic acid to acetic acid and 1,2-propanediol by *Lactobacillus buchneri*. Appl. Environ. Microbiol. 67, 125–132.

Ge, X., Matsumoto, T., Keith, L., Li, Y., 2014. Fungal pretreatment of albizia chips for enhanced biogas production by solid-state anaerobic digestion. Energy Fuels 29, 200–204.

Ge, X., Xu, F., Vasco-Correa, J., 2015. Giant reed: a competitive energy crop in comparison with miscanthus. Renewable Sustainable Energy Rev. 54, 350–362.

Hattingh, W.H.J., Thiel, P.G., Siebert, M.L., 1967. Determination of protein content of anaerobic digesting sludge. Water Res. 1, 185–189.

Herrmann, C., Heiermann, M., Idler, C., 2011. Effects of ensiling, silage additives and storage period on methane formation of biogas crops. Bioresour. Technol. 102, 5153–5161.

Kreuger, E., Nges, I.A., Björnsson, L., 2011. Ensiling of crops for biogas production: effects on methane yield and total solids determination. Biotechnol. Biofuels 4, 44.

Liu, S., Ge, X., Liew, L.N., Liu, Z., Li, Y., 2015. Effect of urea addition on giant reed ensilage and subsequent methane production by anaerobic digestion. Bioresour. Technol. 192, 682–688.

Oleskowicz-Popiel, P., Thomsen, A.B., Schmidt, J.E., 2011. Ensiling-wet-storage method for lignocellulosic biomass for bioethanol production. Biomass Bioenergy 35, 2087–2092.

Pakarinen, A., Maijala, P., Jaakkola, S., Stoddard, F.L., Kymäläinen, M., Viikari, L., 2011. Evaluation of preservation methods for improving biogas production and enzymatic conversion yields of annual crops. Biotechnol. Biofuels 4, 1–13.

Selig, M., Weiss, N., Ji, Y., 2008. Enzymatic Saccharification of Lignocellulosic Biomass, National Renewable Energy Laboratory Technical Report NREL/TP-510-42629.

Shinners, K.J., Binversie, B.N., Muck, R.E., Weimer, P., 2007. Comparison of wet and dry corn stover harvest and storage. Biomass Bioenergy 31, 211–221.

Sluiter, A., Hames, B., Ruiz, R., Scarlata, C., Sluiter, J., Templeton, D., Crocker, D., 2008. Determination of structural carbohydrates and lignin in biomass, National Renewable Energy Laboratory Technical Report NREL/TP-510-42618.

Sultana, A., Kumar, A., 2011. Optimal configuration and combination of multiple lignocellulosic biomass feedstocks delivery to a biorefinery. Bioresour. Technol. 102, 9947–9956.

Tekere, M., Zvauya, R., Read, J.S., 2001. Ligninolytic enzyme production in selected sub-tropical white rot fungi under different culture conditions. J. Basic Microbiol. 41, 115–129.

Vasco-Correa, J., Li, Y., 2015. Solid-state anaerobic digestion of fungal pretreated *Miscanthus sinensis* harvested in two different seasons. Bioresour. Technol. 185, 211–217.

Wahid, R., Nielsen, S.F., Hernandez, V.M., Ward, A.J., Gislum, R., JØrgensen, U., MØller, H.B., 2015. Methane production potential from *Miscanthus* sp.: effect of harvesting time, genotypes and plant fractions. Biosyst. Eng. 133, 71–80.

Wan, C., Li, Y., 2010. Microbial pretreatment of corn stover with *Ceriporiopsis subvermispora* for enzymatic hydrolysis and ethanol production. Bioresour. Technol. 101, 6398–6403.

Wan, C., Li, Y., 2011a. Effect of hot water extraction and liquid hot water pretreatment on the fungal degradation of biomass feedstocks. Bioresour. Technol. 102, 9788–9793.

Wan, C., Li, Y., 2011b. Effectiveness of microbial pretreatment by *Ceriporiopsis subvermispora* on different biomass feedstocks. Bioresour. Technol. 102, 7507–7512.

Weiland, P., 2010. Biogas production: current state and perspectives. Appl. Microbiol. Biotechnol. 85, 849–860.

World Energy Council, 2013. World Energy Resources: Bioenergy. World Energy Council. URL: https://www.worldenergy.org/wp-content/uploads/2013/10/WER_2013_7_Bioenergy.pdf. (accessed on 25.07.2015).

Xu, F., Wang, Z.W., Li, Y., 2014. Predicting the methane yield of lignocellulosic biomass in mesophilic solid-state anaerobic digestion based on feedstock characteristics and process parameters. Bioresour. Technol. 173, 168–176.

Yahaya, M.S., Kimura, A., Harai, J., Nguyen, H.V., Kawai, M., Takahashi, J., Matsuoka, S., 2001. Effect of length of ensiling on silo degradation and digestibility of structural carbohydrates of lucerne and orchardgrass. Anim. Feed Sci. Technol. 92, 141–148.

Yang, H.Y., Wang, X.F., Liu, J.B., Gao, L.J., Ishii, M., Igarashi, Y., Cui, Z.J., 2006. Effects of water-soluble carbohydrate content on silage fermentation of wheat straw. J. Biosci. Bioeng. 101, 232–237.

Yang, L., Li, Y., 2014. Anaerobic digestion of giant reed for methane production. Bioresour. Technol. 171, 233–239.

Zeng, M., Mosier, N.S., Huang, C.-P., Sherman, D.M., Ladisch, M.R., 2007. Microscopic examination of changes of plant cell structure in corn stover due to hot water pretreatment and enzymatic hydrolysis. Biotechnol. Bioeng. 97, 265–278.

Zhao, J., Ge, X., Vasco-Correa, J., Li, Y., 2014. Fungal pretreatment of unsterilized yard trimmings for enhanced methane production by solid-state anaerobic digestion. Bioresour. Technol. 158, 248–252.

Zheng, Y., Yu, C., Cheng, Y.S., Zhang, R., Jenkins, B., VanderGheynst, J.S., 2011. Effects of ensilage on storage and enzymatic degradability of sugar beet pulp. Bioresour. Technol. 102, 1489–1495.

Zheng, Y., Zhao, J., Xu, F., Li, Y., 2014. Pretreatment of lignocellulosic biomass for enhanced biogas production. Prog. Energy Combust. Sci. 42, 35–53.